



Exploiting Saliency in Attention Based Convolutional Neural Network for Classification of Vertical Root Fractures

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Abstract. Cone-beam computed tomography (CBCT) is widely used in clinical diagnosis of vertical root fractures (VRFs) which presents as crack on the teeth. However, manually checking the VRFs from a larger number of CBCT images is time-consuming and error-prone. Although the Convolutional Neural Networks (CNN) have achieved unprecedented progress in natural image recognition, end-to-end CNN is unsuitable to identify VRFs due to crack appears to be multi-scales and their complex relationships with surroundings tissues. We proposed a novel Feature Pyramids Attention Convolutional Neural Network (FPA-CNN), which incorporates saliency mask and multi-scale feature to boost the classification performance. Saliency map is viewed as spatial probability map where a person might look first to make a discriminative conclusion. Therefore it plays a role of high-level hint to guide the network focusing on the discriminative region. Experimental results demonstrate that our proposed FPA-CNN overcomes the challenge arisen from multi-scale crack and complex contextual relationships.

Keywords: Vertical root fractures · Attention · Weakly supervised

1 Introduction

Cone-beam computed tomography (CBCT) is the most sensitive imaging technique with the lowest medical exposure in diagnosing incomplete vertical root fracture (VRFs) among the common radiological examinations [19]. The early detection of VRFs is important to prevent extensive and additional damages to the periodontal tissues and also unnecessary treatment and costs [25]. Unfortunately, clinical examination provides limited information for VRF detection and is based on unspecific signs and symptoms [3]. The radiologic evaluation of VRFs may be improved by the use of tridimensional imaging such as cone-beam computed tomographic (CBCT) imaging [4, 21].

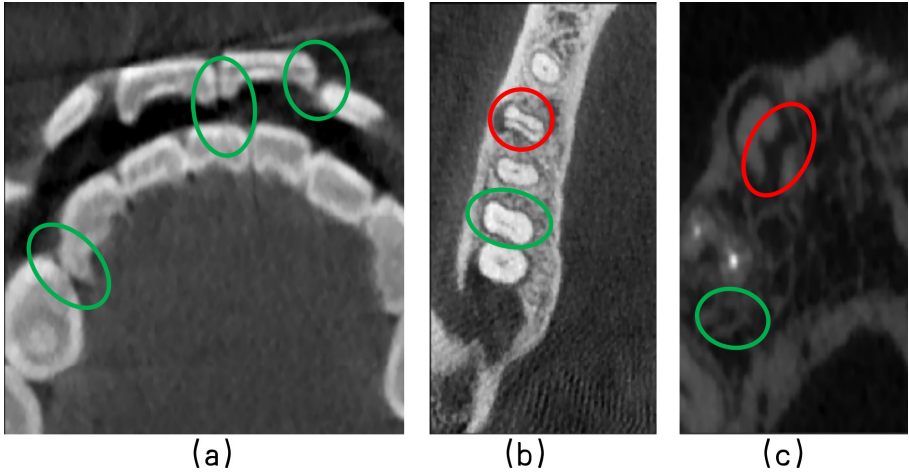


Fig. 1. The CBCT image samples. The co-occurrences of multi-scale crack in different regions makes the classification difficult to CNNs. The red circles denote the VRFs region, while the green circles denote the healthy tissue that is prone to be identified as VRFs. (a) is located in cross area between upper and lower jaw, which shows that there are many cracks which should not be identified as VRFs. (b) and (c) show that there exist confused cracks in various scales. (Color figure online)

Manually checking the existence of the VRFs from CBCT images one by one is time-consuming and error-prone. Hence, it is practically meaningful to develop an automatic method for checking the VRFs via CBCT images. Deep Convolutional Neural Network (CNN) is a special type of Neural Networks. CNN have proven to be useful models for tackling a wide range of visual tasks such as classification [13, 30], Segmentation [20, 26] and Objection Detection [24]. The powerful learning capability of CNN is due to the use of hierarchical feature extraction layers which obtain semantics from input images.

The challenge of identifying VRFs in CBCT images is arised from multi-scale cracks and complex contextual relationships. As shown in Fig. 1. Crack in neighbouring teeth and rows of teeth is similar to the real VRFs in a range of scales. Therefore, high spatial resolution and strong semantic information are required to achieve this challenge. However, the spatial resolution and semantic information compete with each other. Hence, the traditional end-to-end CNN is unsuitable for this problem.

Most successful CNNs perform outstandingly under the supervised learning framework, which require ground-truth labels to be given for a big training data. The pixel-level, region-level and image-level ground-truth are fed into the CNNs to process Segmentation, Detection, and classification model building respectively. Recently, weakly supervised learning has made big progress. These works use low-level ground-truth at train stage, and predict the high-level label, e.g. Additional informations such priori knowledge, reasonable constraint, and

partial labels are exploited for achieving the goal. Among these researches about weakly supervised learning, many works indicate that CNN trained for classification using image-level label has the power of locating. This means CNN implicitly learns how to extract the discriminate region for classification task. This is also the reason why these weakly supervised learning CNN should initialize from a well trained classification CNN instead of initializing it from scratch. As mentioned above, the classification CNN designed for nature image recognition can not work normally on this problem. Well trained classification CNN implies high-level knowledge. In reverse, can some high-level hint help training a good classification CNN?

In this paper, we exploit the coarse image saliency in guiding the CNN to learning the feature of VRFs. We build some coarse image saliency in which there are pixels labelled as seed. After that, a level-set method are applied to generate a more reasonable image saliency from seed pixels. Then, these image saliency actor as high-level hint, which guides the region based attention module to learning the saliency features. Finally, the saliency is used as soft attention probability mask to let the classification submodule focusing on discriminative regions. All in all, We proposed a Feature Pyramids Attention Convolutional Neural Network (FPA-CNN) which incorporates saliency mask to boost the classification performance in our challenging dataset.

The rest of paper is organized as follows. We first review some related works, and clarify the difference from ours in Sect. 2. And then we introduce database used in our study, later present the FPA-CNN in Sect. 3. In Sect. 4, we evaluate the performance of our method by comparing with several state-of-the-art CNN architecture.

2 Relate Work

2.1 VRFs Recognition

In literature, the neural network is introduced for detecting vertical root fracture in ex-Vivo [12]. They use extracted premolars with no carious lesions as study object and the adjacent anatomic structures are ignored. Johari [10] takes almost the same routine with additional denoising method. The main difference between their work and ours are that all the CBCT images in our dataset are captured in-vivo. So our images contains the whole jaw and are affected by artifacts. What's more, the difference between individuals can not be ignored. In contrast to FPA-CNN, their work focus on images which contains only one tooth. As far as we know, these has not been exploiting by preview works.

2.2 CNN Based Image Classification

Starting with LeNet-5 [14] and AlexNet [13], convolutional neural networks (CNN) have typically a standard structure, stacked convolutional layers are followed by one or more fully-connected layers. There are many state-of-the-art

CNN perform well on nature images. [28] and [6] boost the CNN classification capability using deeper hierarchical layers and the shortcut structures. [9, 29, 30] make a tradeoff on width and depth of the network. And features from various receptive field are fused to get better discriminate feature. Hu et al. [7] propose a mechanism that allows the network to perform feature recalibration, which can learn global information to selectively emphasise informative features and suppress less useful ones. Huang et al. [8] connects each layer to every other layer in a feed-forward fashion, which result in significant improvements at the cost of more graphic memory at training stage.

Although large scaled CNN requires a great amount of data, we can transfer the CNN pretrained on large dataset to a domain which has a small amount of data. Our FPA-CNN use a pretrained resnet50 as our basic networks to extract the shared features for next feature pyramids module and region based attention module.

2.3 Weakly Supervised Learning

Weakly Supervised Learning in CNN. Weakly supervised learning is an umbrella term covering a variety of studies that attempt to construct predictive models by learning with weak supervision [34]. Here we focus on a specified situation where region-level or pixel-level information are implicitly learnt using image-level labels. Zhou et al. [33] indicate classification CNN is able to localize the discriminative image regions on a variety of tasks despite not being trained for them. This localization ability can even be improved by using global average pooling layer, global max pooling layer [32], or log-sum-exp pooling layer [23]. Recently, Zhu et al. [35] designs a network module, Soft Proposal(SP), which can be plugged into any standard convolutional architecture to introduce the nearly cost-free object proposal.

This sounds free ability of classification CNN make many weakly supervised learning possible. Ahn et al. [1] learning instance segmentation with image-level class labels as supervision. They combine the localization ability of classification CNN and traditional image processing methods to generates pseudo instance segmentation labels of training images. Then the pseudo instance segmentation are used to train a fully supervised model. Joon et al. [22] add extra information called ‘seed’ into the processing of generating pseudo. Furthermore they use the region proposed by well trained classification CNN as a binary mask segmenting the one object a person is most likely to look first. Image saliency has multiple connotations. Binary mask is just one kind of definition. In this paper, we view image saliency as a spatial probability map of where a person might look first to make a discriminative conclusion, which is similar to [31].

Propagating Labels. Inspired by [31], we define image saliency as a spatial probability map of most discriminative region. The saliency map can be generated by classification CNN in a weakly supervised method mentioned as above, or it can also be labeled by human which is also called ‘seed’. One simple way

to propagate the label from known pixel to unknown pixel is to merge the pixel into super pixel. This process is very similar with clustering algorithm, and the ‘distance’ is defined as the similarity between pixels. [2, 5, 27] can take full use of neighbouring information around the pixel to group pixels. There are many works combining the CNN and information propagated on image saliency [16, 22, 35]. However these works only take the information around the label into account.

In this paper we utilize the active contour model [11], also called snakes to propagate the information from the local seed to the whole image, which considers the global information in image. Early active contour models are formulated in terms of a dynamic parametric contour $C(s, t) : [0, 1] \times [0, \infty] \rightarrow R^2$ with a spatial parameter s in $[0, 1]$ which parameterizes the points in the contour, and a temporal variable $t \in [0, \infty]$. The active contour model given in a level set formulation is called an implicit active contour or geometric active contour model, which is able to handle topological changes, such as splitting and merging, in a natural and efficient way. So we take distance regularized level set (DRLS) [15] method as our information propagation method. The details will be explained in next section.

3 Materials and Method

3.1 VRFs DataSets

The data is collected randomly from patients in cooperative stomatological hospital, which guaranteeing the practical usage of final models driven by it. The data desensitization is also applied for privacy protection. The dataset was consist of 9219 CBCT images coming from 37 subjects. There are about 250 images per subject, which generated by scanning in vertical directions. Among those, there are only 1868 positive instance, in which there exist VRFs. By the time we trained the networks, and is growing bigger. We aim to label at least 5k positive instance regardless of the negative quantity. Patients are in sitting position and keep the mandible parallel with ground during the scanning. All image is collected by a NewTom VG with 110kv of X-ray tube voltage, and 0.125mm of reconstructed resolution. Traditional techniques of image enhancement such as Gaussian filter and histogram equalization are applied to highlight the fractures for next labelling work.

At the labelling stage, dentists label each image that exists VRFs as positive instance. What’s more, there are about 18 subjects have the scribble on the teeth. These scribble indicating the image saliency, i.e. the spatial probability map of most discriminative region. Most important of all, these scribble is easily obtained and can be done by non-professional person. Three typical types of weak supervision [34]: incomplete supervision, where only a subset of training data is given with labels; inexact supervision, where the training data are given with only coarse-grained labels; and inaccurate supervision, where the given labels are not always ground-truth. Scribble labels used in our datasets play a role of incomplete supervision. That means the scribble only indicating the partial discriminative region, so the level-set is applied to propagate the information

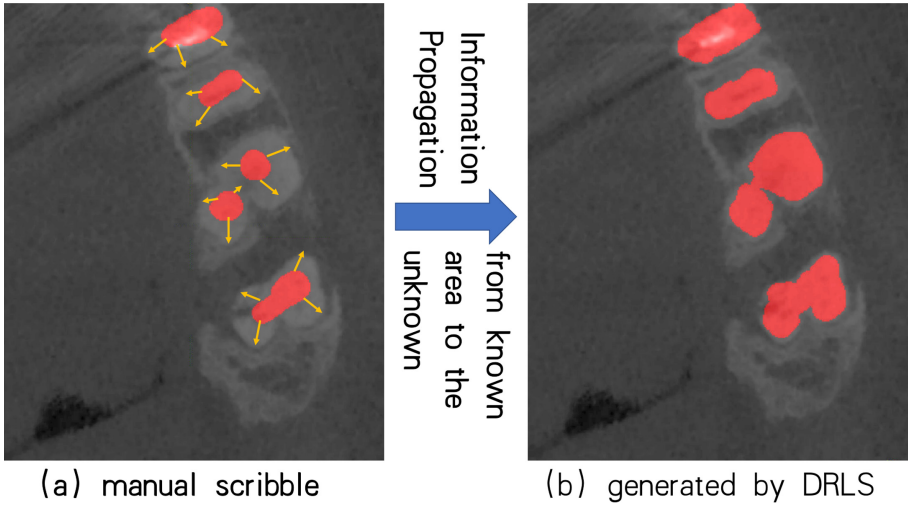


Fig. 2. The information propagation and subjects difference. (a) the red regions are labeled manually, indicating the discriminative regions. However, only partial of regions are labeled. So the level-set are applied to propagate the information from the known regions to unknowns. The yellow arrows show the directions of information propagation (b) shows the results generate image saliency. Notice that all the teeth have been marked by the generate labels, which will guide the network to checking the existence of VRFs. Comparing to Fig. 1 this fig also shows the differences between the subjects (The contrast of images from different subject varies), which increase the difficulty of VRFs recognition. (Color figure online)

from local to global. What's more, there are only 50% of images have scribbled labels, which can also be viewed as incomplete supervision. So regularized level set is applied to propagate information from the partial labels to global images. These generated saliency plays a role as high-level hint in our FPA-CNN to guides the networks choose the right discriminative regions. Figure 2 shows this process of the labeling and the generated saliency map in our dataset.

3.2 Feature Pyramids Attention Convolutional Neural Network

In order to incorporate the generated saliency and recognize the various scales of VRFs. We propose a novel classification network FPA-CNN. There are two important modules. One of which is feature pyramids module, the other is soft attention module. ResNets [6] are adopted as our backbone, on which the modules are attached. The FPA-CNN structure can be seen in Fig. 3.

Feature Pyramids Module. The feature pyramids [17] are a basic module in recognition systems for detecting objects at different scales. As mentioned above, The co-occurrences of crack in different scales requires network has well

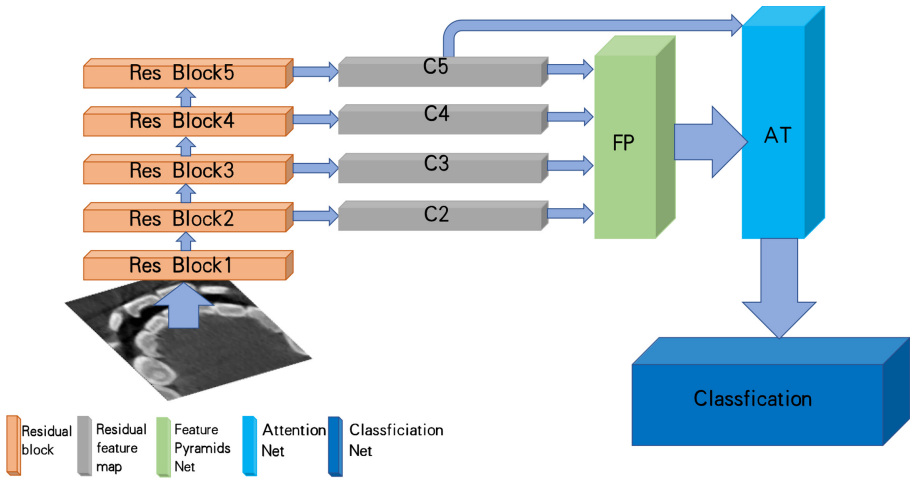


Fig. 3. FPN-CNN overview. ResNets are adopted as backbone, on which the feature pyramids module and attention module are attached. At last the processed feature maps are fed into classification net.

perception in multi spatial resolution. So we add Feature pyramids module to our backbone network to get a feature extractor that can handle the various scales of VRFs.

There are two significant information flow in feature pyramids. The bottom-up pathway is the feedforward computation of the backbone ConvNet, which computes a feature hierarchy. Specifically, for ResNets there five residual block. Denoting the output of these residual blocks as C_1, C_2, C_3, C_4, C_5 for conv1, conv2, conv3, conv4, and conv5 outputs, which have strides of 2, 4, 8, 16, 32 pixels with respect to the input image. However, only C_2, C_3, C_4, C_5 are used to generate the final feature maps, in consideration of graphic memory (C_1 's feature map consumes a lot).

The topdown pathway fuses the low-level features and high-level features. The low-level features have strong spatially resolution, while the high-level features have strong semantically information. The high-level feature maps is then merged with the corresponding low-level map (which undergoes a 1×1 convolutional layer to reduce channel dimensions) by upsampling followed by element-wise addition. Additional 3×3 convolution is applied on each merged map to generate the final feature map set P_2, P_3, P_4, P_5 . This set of fusion feature maps makes perception in multi resolution possible. The details of feature pyramids module is shown in Fig. 4.

Saliency Driven Attention. VRFs can be view as cracks in teeth. In the mean while, there are a large number of regions seem look crack, e.g. the regions between teeth and healthy tissue (green circles in Fig. 1 picture (b)) Attention module helps later classifier overcoming these confusing regions.

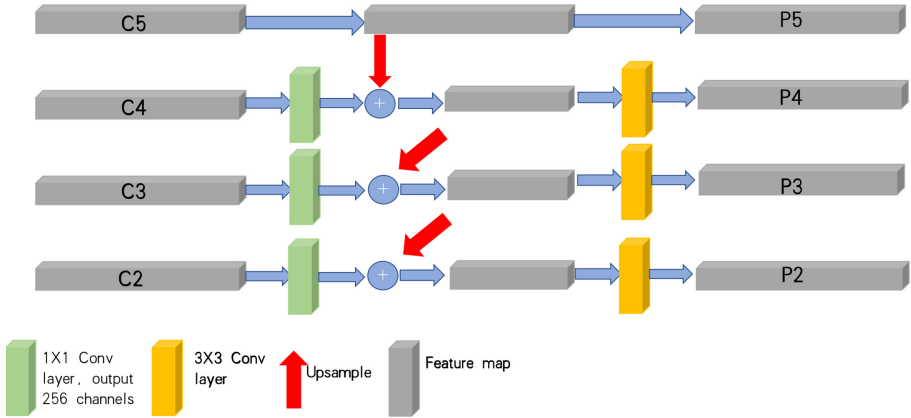


Fig. 4. Feature pyramids module. The fusion feature maps makes perception in multi resolution possible.

Attention module is actually a small network. It uses the feature maps C_5 as its input features. Its output is spatial probability map of where a person might look first to make a discriminative conclusion, that is the image saliency. C_5 undergoes a 3×3 convolutional layer to get the context information around the pixels, then followed by a 1×1 convolutional layer to reduce channel dimensions.

The final outputs has two channels presenting the degree of confidence that the region should be a discriminative area. A spatial probability map A_1 is obtained by Softmax layer regularizing the outputs. This A_1 indicates the most discriminative regions on which the classification decision should be made.

Features Aggregation. The classification subnet combines the outputs from feature pyramids module and attention module. The attention output A_1 is applied to each features in P_2, P_3, P_4, P_5 by element-wise multiplication. Upsampling is applied if the size of two features mismatch in shape. In A_1 Those discriminative regions have a higher weights, so the multiplication of A_1 and P_n where $n \in 2, 3, 4, 5$ is actually a soft attention which emphasize the focus regions and reserve the less importance regions. And we denote the results of multiplication as P'_2, P'_3, P'_4, P'_5 . Then a ROI max pooling [24], also called adaptive max pooling, is applied to each feature map in P'_2, P'_3, P'_4, P'_5 to get the same features shape in regardless of inputs size. Finally, The results of pooling are concatenated and flatted before sent to the full connected layer. This is shown in Fig. 5.

Joint Training. FPA-CNN has two kinds of loss items. Attention loss measures difference between the generated discriminative regions from level-set and those from attention net. Due to there are only 50% images have supervision information from level-set method, the attention loss is optional. Classification loss is the main loss, which guides the FPA-CNN to learning the proper information.

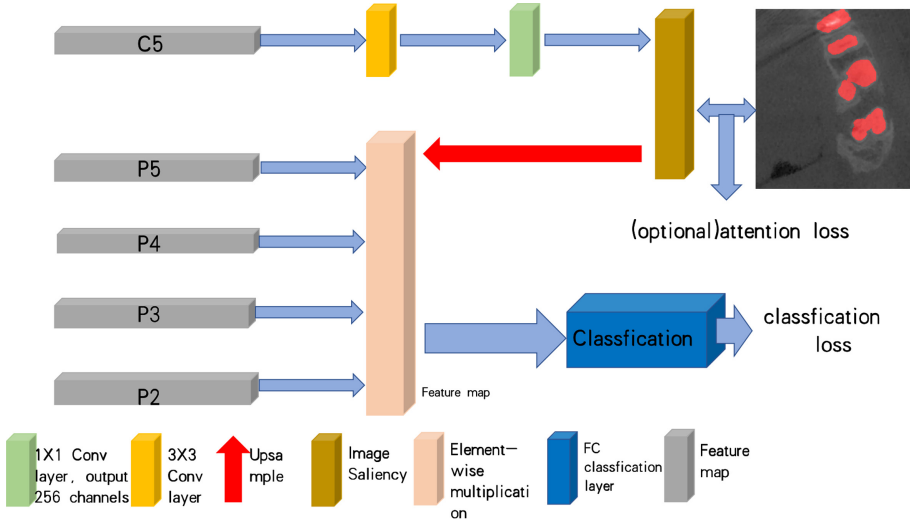


Fig. 5. Attention module and classification. The attention module generates the image saliency, which is used on the inputs of classification net as soft attention.

What's more, the classification loss can affect the attention module, thus the generated image saliency mainly incorporates the classification information and the supervision image saliency for attention module only play a role of assistant at the very begin of training.

4 Experiments

we compare our FPA-CNN with ResNets [6], InceptionV3 [30] by calculating the Area Under Curve (AUC) of Precision-Recall (PR) Curve. The reason why we choose PR curve is that we only care about identifying the VRFs. Even if the image do not contains the VRFs, there is no promise that other symptom do not exist. Therefor PR curve is a good choice due to it focus itself on single class. First of all, we present some experimental setting. The learning rate and weights decay are 10^{-5} and 10^{-2} respectively for next experiments. Stochastic Gradient Descent with momentum 0.9 is applied as our optimizer. What's more, we train each networks until the classification loss value on testing dataset doesn't descend for 5 epoch to avoid over-fitting. As for the spliting of dataset, we split the dataset into training part and testing part in two way. The one is splitting at image level which is a general way in nature image domain, the other is splitting at subject level which considers the differences between subjects and the image similarity from same subject. Due to the images from the same subject are actually continuous in spatially, splitting at image level will implicitly introduce the information of training dataset to testing. When there comes a new subject, the network fail. Splitting at subject level is suitable for this special dataset.

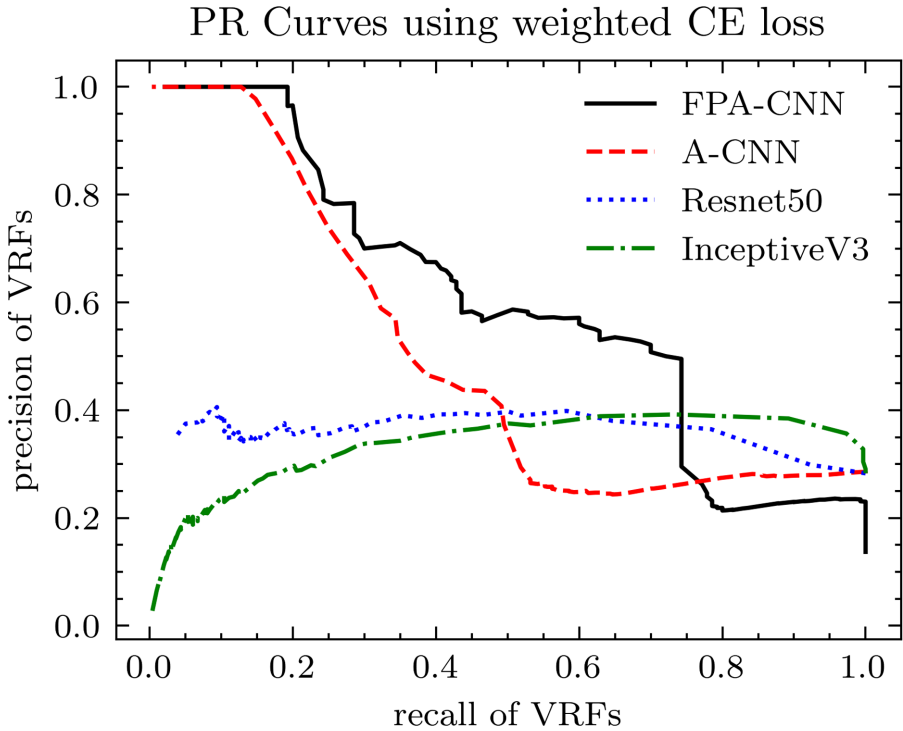


Fig. 6. Network Precision-Recall curve

In the next experiments, we pick 33 subjects from 37 as training data and the remains used as testing data.

Another key points is that dataset is unbalance, so the image containing the VRFs is in the minority. In order to elimination arised from the imbalance data, we choose the weighted cross entropy and focal loss [18] as our classification loss. The positive and negative have the coefficients of 0.9 and 0.1 respectively in weighted cross entropy. And the hyperparameter γ in focal loss is 2, which is a relatively moderate. Table 1 shows this two kinds of loss function. We observe that the weighted cross entropy performs better than focal loss, therefore we adopt the weighted CE loss as out standard function loss for next experiments.

Figure 7 shows the network PR curve on test dataset. A-CNN's structure is the similar to FPA-CNN except features pyramids module. Although Resnet50 and InceptionV3 seem to have a little discriminative capability, the 26% is almost reaching the positive rate in test dataset. This result indicates that Resnet50 and InceptionV3 has almost no discriminative capability on test dataset. A-CNN makes a big progress with the help of region proposal attention module. However this discriminative capability stay in a narrow interval, and there is a steep drop in PR curve. The FPA-CNN gain a relative wider range of discriminative probability, due to the feature pyramids module provides a multi-scale adaptive

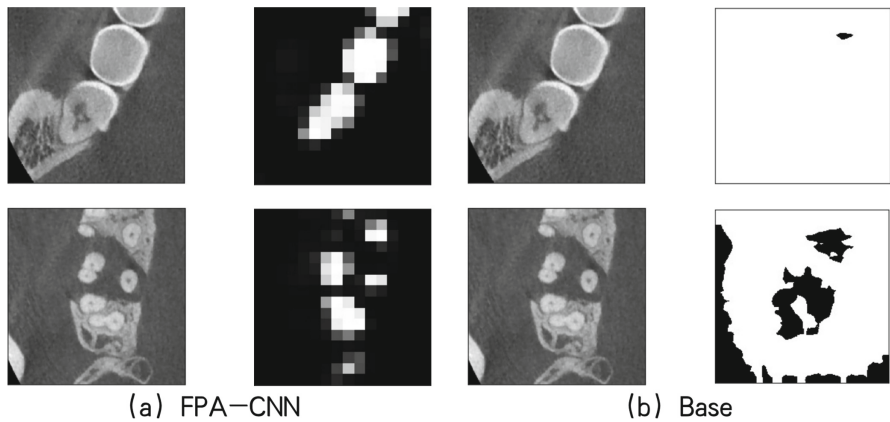


Fig. 7. Attention regions for classification decision

features. Then we compare the focusing regions of network, Fig. 6 shows the saliency image of discrimination, from which we find a possible explanation of the poor performance of Resnet50 and InceptionV3. In contrast to original version, our FPA-CNN focus on the suitable regions which enhances the useful features as well as reduces the misleading information introduced by backgrounds.

Table 1. The AUC results on different loss function

Method	Base	Focal Loss	Weighted CE
Resnet50	0.327	0.332	0.365
InceptionV3	0.315	0.323	0.334
FPA-CNN	0.565	0.573	0.592

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